

Chapter 1

Project Proposal

Multiplex Graph Attention Network Using Gene Co-expression Networks for Disease Prediction

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1.1 Introduction and Description

Multiple sclerosis (MS) is a complex, polygenic autoimmune disorder wherein the immune system attacks myelin, the protective cover of nerve cells (axons) in the central nervous system (CNS)[5]. Although damage can be partially mitigated by the re-routing of messages using undamaged nerve cells (plasticity), the damage eventually becomes too extensive to compensate for it, resulting in progressive neurodegeneration. A multitude of clinical tests are currently used in diagnosing MS, and recently, AI has shown promise in providing more personalised treatments, disease monitoring and the identification of biomarkers that may aid in disease prognosis and diagnosis [2], the lattermost being the focus of this project.

Gene expression is the process outlined by the central dogma of molecular biology: DNA is transcribed into RNA, and RNA is translated into functional products. The measure of how much a gene is expressed at a given time is referred to as its gene expression level, and the analysis of differential expression levels when comparing healthy people and diagnosed patients can aid in the development of disease signatures based on changes in gene activity.

Transcriptomics is a sub-branch of omics focused on the study and analysis of the transcriptome, the collection of all RNA transcripts present in a sample at a given time. Epigenomics is the study of the chemical compounds that control the behaviour of the genome. Studying the epigenome may in future be used to detect harmful molecular changes before physical symptoms appear, acting as an early warning signal[1, 8].

Single-cell RNA-sequencing as described in Luecken and Theis is a technique used to produce raw

RNA sequence reads which can be used to produce gene expression matrices, where rows represent genes and each column represents a single cell, with a matrix cell representing how active that particular gene is. Analogously, scATAC-seq analyse open chromatin (which allow for gene expression) to produce peak count matrices where rows represent peaks and columns represent individual cells, with the individual matrix cells representing the number of sequence reads. Advanced, high-throughput technologies for studying gene expression, such as single-cell RNA sequencing(scRNA-seq) and single-nucleus RNA sequencing (snRNA-seq), allow gene expression to be measured at the level of individual cells or nuclei, enabling the heterogeneity of cell populations to be explored. Studies have utilised snRNA-seq data in clustering distinct MS sub-groups[12] providing a basis for use in future individualised treatment.

Given that the complexity of biological networks in higher organisms primarily arises from dynamic gene interactions as opposed to the size of the genome (C-value paradox), analysing gene dynamics offers a more informative framework for understanding biological systems. Gene-gene co-expression networks (GGCNs) are typically presented as undirected graphs where nodes are genes and the presence of edges depends on co-expression values calculated from gene expression data[17]. These networks can be used to identify the functional status of genes from a systemic perspective, which can in turn inform the discovery of gene-disease associations through guilt-by-association analyses. Several frameworks have been used to infer GGCNs from snRNA-seq data, including the WGCNA framework, and its extension hdWGCNA, which will be used to build the multi-layer co-expression network for this project.

Graph neural networks (GNNs) have commonly been used for tasks like graph classification, node classification and link prediction[15]. Given a graph $G(V, E)$ as input in the form of a node feature matrix ($H : \mathbb{R}^{N \times F}$), GNNs produce new node embeddings that capture the node features, and the learnt embeddings can be used for graph classification. Graph Attention Networks (GATs) introduce a masked attention mechanism which works by constraining the nodes taken into account in computing the updated features for a node to its neighbourhood[18] – previous works have successfully applied these networks to disease prediction[13]. To reflect the cell-type specific gene co-expression, a multi-layer graph, known as a multiplex graphs will be used, requiring the use of a multiplex GAT (mGAT).

In this project, snRNA-seq data from diagnosed MS patients and healthy control will be transformed into multi-layer co-expression networks using the hdWGCNA framework. These graphs will be used as input to an mGAT whose initial downstream task will be binary MS classification. The contributions of the project are:

- The leveraging of the hdWGCNA framework to construct the multiplex co-expression networks
- The design of an mGAT architecture that uses the GGCNs as input for downstream binary classification
- The performance comparison of the proposed mGAT architecture against the baseline GAT model.

1.2 Substance and Structure

1.2.1 Datasets

The datasets to be used for this project are items [3, 6, 11, 14] and each include a subset of transcriptomic (snRNA-seq) and epigenomic (snATAC-seq) data.

1.2.2 Data Processing and Libraries

hdWGCNA is an extension of the WGCNA framework which allowed bulk RNA-seq data to be transformed into gene co-expression networks adapted to handle higher dimensional single-cell datasets. Given the output of cell-type specific GGCNs, a multiplex graph can be constructed from the outputs. A multiplex graph is informally defined as a graph where nodes are split into layers and both interlayer and intralayer edges may be present.

The choice of a graph architecture reflecting the heterogeneity of cells in samples is due to the differential expression levels of different cell types, however, whether this affects classification will be tested by the inclusion of a traditional GAT. Whilst initially written in R, a Python re-implementation, py-hdWGCNA, has been written that is numerically faithful to the original package, and will be used.

The tensor data structure and associated operations from the PyTorch library will be used to write the linear layers and graph attention layers as required for the GAT, mGAT and MLP models rather than using the prebuilt library functions. The sci-kit library will be used to split the processed dataset into training, validation and testing datasets.

1.3 Models

The GAT architecture must be adapted to handle multiplex graphs: the exact network architecture (topology, hyperparameters and layers) is to be designed, it will modify previous implementations of multiplex GATs, namely [4] for graph classification.

The choice of baseline models for comparisons are inspired by Luzhnica, Day, and Lio. The rationale being on the basis that the MLP model provides a graph structure-blind baseline to assess whether incorporating the graph structure improves performance and the standard GAT allows the assessment of the informativeness of a cell-type specific graph.

1.4 Starting Point

I have some experience with machine learning beyond the Computer Science Tripos, however I was unfamiliar with graph attention networks, so I have done some preliminary reading. In particular, I have read [13, 16, 18, 19] however further reading will be required as noted in the timetable.

1.5 Success Criterion and Evaluation

The project will be considered a success if the following criteria are met:

- MLP, mGAT and GAT models are designed and implemented for graph classification.
- Real-world transcriptomic data is used to train the model.
- The evaluation framework is carried out on the models to assess their performance.

The evaluation framework for the project will use a confusion matrix, allowing for 4 classification metrics to be calculated: accuracy, precision, recall and a recall-oriented F-score given that missing a diagnosis of MS may delay medical intervention.

1.6 Extensions and Further Work

If the success criteria are met, I will attempt to implement the following extensions:

- EX1: Implement a multi-class classification model using transcriptomic data from Alzheimers' and ALS patients and evaluate with a new evaluation framework.
- EX2: Incorporate epigenomic (snATAC-seq) data to produce a multi-omic classification model.
- EX3: Improve the explainability of the model using the DEGREE framework[7] and evaluate with fidelity tests.

1.7 Work Plan

Michaelmas Term

- **13/10/26–26/10/26**
Background reading on mGATs, GATs and the hdWGCNA package and set up code repository.
Milestone: Production of a report explaining the theory about these model and the reason for their inclusion in the project.
- **(27/10/26–9/11/26)**
Read up on PyTorch and decide on the network architecture for the three models.
Milestone: Initial diagrams for each network architecture and produce basic ML examples in PyTorch.
- **10/11/26–23/11/26**
Process the snRNA-seq dataset using the py-hdWGCNA framework and split data into training, validation and testing datasets at a patient level.
Milestone: Datasets should be ready for training.
- **24/11/26–7/12/26**
Implement the mGAT, GAT and MLP models, using the training dataset.
Milestone: 3 working models which are able to perform the classification task.

Winter Break

- **8/12/26–21/12/26**
Hyperparameter optimisation and evaluation of model on test dataset.
Milestone: Production of written overview of the evaluation results as outlined in “Success Criterion and Evaluation”
- **22/12/27–4/1/27**
Prepare the progress report and ensure the core success criteria have been met.
Milestone: Progress report is written, prepare for the presentation.
- **5/1/27–18/1/27**
Do the additional research for extensions [1-3]

Milestone: Production of document about multimodal fusion, determining when modal fusion will be implemented, identifying evaluation metrics for multi-class classification and detailing the explainability mechanisms to be implemented.

Lent Term

- **19/1/27–1/2/27**

Begin work on extension 1.

Milestone: Catching up on any missed milestones.

- **2/2/27–15/2/27**

Friday 5 February 12pm: Progress Report Presentation Deadline

Finish work on extension 1, start Introduction and Preparation chapters, work on extension 2.

Milestone: Working multi-class classification model.

- **16/2/27–1/3/27**

Friday 26 February: Progress Report Presentations

Continue extension 2, begin writing Implementation Chapter.

Milestone: Send Introduction and Preparation chapters for review.

- **2/3/27–15/3/27**

Finish Implementation Chapter and start on Evaluation and Conclusion chapters, start extension 3 time permitting.

Milestone: Implementation chapter sent for review and working implementation of a multi-omic multi-class classification system.

Easter Break

- **16/3/27–29/3/27**

Complete Evaluation and Conclusion, and finish extension 3.

Milestone: First draft of dissertation sent for review, production of report exploring feature importance using the chosen framework.

- **30/3/27–12/4/27**

Buffer period, edit dissertation based on feedback received.

Milestone: Second draft of dissertation sent for review.

- **13/4/27–26/4/27**

Format and proofread final draft of dissertation.

Milestone: Submission of dissertation.

Easter Term

- **27/4/27–9/5/27**

Friday 11 June 12pm: Dissertation Deadline

1.8 Resource Declaration

I will be using my personal laptop, a 2024 Dell Inspiron 16 2-in-1, with 16GB memory and a 1TB SSD for software development. As a backup, I will use my secondary personal laptop with my project backed onto my university account OneDrive. I will continuously backup my code and dissertation with GitHub version control. I accept full responsibility for this machine and I have made contingency plans to protect myself against hardware and/or software failure. In terms of software, I will use Python including the PyTorch and py-hdWGCNA libraries.

The project also requires the use of biological datasets for training data. These datasets have been taken from <https://zenodo.org/records> which are publicly available to anyone and have been downloaded onto my personal device. Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) was accessed on 18/09/2026 from <https://registry.opendata.aws/allen-sea-ad-atlas>.

Bibliography

- [1] Tomás Alarcón, Javier A. Menendez, and Josep Sardanyés. *Metabolic Early Warning Signals of Epigenetic Tipping Points under Chromatin Modifier Competition*. Mar. 25, 2025. DOI: 10.1101/2025.03.24.644914. URL: <http://biorxiv.org/lookup/doi/10.1101/2025.03.24.644914> (visited on 09/18/2026). Pre-published.
- [2] Moein Amin et al. “Artificial Intelligence and Multiple Sclerosis”. In: *Current Neurology and Neuroscience Reports* 24.8 (Aug. 2024), pp. 233–243. ISSN: 1528-4042, 1534-6293. DOI: 10.1007/s11910-024-01354-x. URL: <https://link.springer.com/10.1007/s11910-024-01354-x> (visited on 09/19/2026).
- [3] N Barretto et al. *A Multi-Omics Characterization of the Human Lumbar Spinal Cord and Motor Cortex Reveals Non-Overlapping Molecular Signatures in ALS [snRNA-Seq; snATAC-Seq]*. URL: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi>.
- [4] Marco Bongiovanni et al. *MPXGAT: An Attention Based Deep Learning Model for Multiplex Graphs Embedding*. Mar. 28, 2024. DOI: 10.48550/arXiv.2403.19246. arXiv: 2403.19246 [cs.LG]. URL: <http://arxiv.org/abs/2403.19246> (visited on 09/19/2026). Pre-published.
- [5] Imane Boutitah-Benyaich et al. “Multiple Sclerosis: Molecular Pathogenesis and Therapeutic Intervention”. In: *Signal Transduction and Targeted Therapy* 10.1 (Oct. 2, 2025), p. 324. ISSN: 2059-3635. DOI: 10.1038/s41392-025-02415-4. URL: <https://www.nature.com/articles/s41392-025-02415-4> (visited on 09/26/2026).
- [6] Maria Louise Elkjaer et al. *Dataset for Article Single-Cell Multi-Omics Map of Cell Type-Specific Mechanistic Drivers of Multiple Sclerosis Lesions*. Zenodo, June 21, 2024. DOI: 10.5281/ZENODO.17733764. URL: <https://zenodo.org/doi/10.5281/zenodo.17733764> (visited on 09/18/2026).
- [7] Qizhang Feng et al. *DEGREE: Decomposition Based Explanation For Graph Neural Networks*. Version 1. 2023. DOI: 10.48550/ARXIV.2305.12895. URL: <https://arxiv.org/abs/2305.12895> (visited on 09/27/2026). Pre-published.
- [8] Barry M Lester, Marie Camerota, and Todd M Everson. “The Emergence of Developmental Behavioral Epigenomics”. In: *Epigenomics* 14.9 (May 2022), pp. 499–502. ISSN: 1750-1911, 1750-192X. DOI: 10.2217/epi-2022-0065. URL: <https://www.tandfonline.com/doi/full/10.2217/epi-2022-0065> (visited on 09/18/2026).
- [9] Malte D Luecken and Fabian J Theis. “Current Best Practices in Single-cell RNA-seq Analysis: A Tutorial”. In: *Molecular Systems Biology* 15.6 (June 1, 2019), MSB188746. ISSN: 1744-4292. DOI: 10.15252/msb.20188746. URL: <https://link.springer.com/10.15252/msb.20188746> (visited on 08/11/2026).
- [10] Enxhell Luzhnica, Ben Day, and Pietro Lio. *On Graph Classification Networks, Datasets and Baselines*. May 29, 2019. DOI: 10.17863/CAM.40372. URL: <https://www.repository.cam.ac.uk/handle/1810/293222> (visited on 09/14/2026). Pre-published.

- [11] Will Macnair and Anna Williams. *Single Nuclei RNAseq Stratifies Multiple Sclerosis Patients into Distinct White Matter Glial Responses*. Zenodo, Sept. 12, 2023. DOI: 10.5281/ZENODO.8338963. URL: <https://zenodo.org/record/8338963> (visited on 09/18/2026).
- [12] Will Macnair et al. *Single Nuclei RNAseq Stratifies Multiple Sclerosis Patients into Distinct White Matter Glial Responses*. Apr. 9, 2022. DOI: 10.1101/2022.04.06.487263. URL: <http://biorxiv.org/lookup/doi/10.1101/2022.04.06.487263> (visited on 08/18/2026). Pre-published.
- [13] Neal G. Ravindra et al. *Disease State Prediction From Single-Cell Data Using Graph Attention Networks*. Version 2. 2020. DOI: 10.48550/ARXIV.2002.07128. URL: <https://arxiv.org/abs/2002.07128> (visited on 09/14/2026). Pre-published.
- [14] *Seattle Alzheimer’s Disease Brain Cell Atlas (SEA-AD)*. URL: <https://registry.opendata.aws/allen-sea-ad-atlas/>.
- [15] Reid C. Sewell. *GNNs for Network Classification in Single Cell RNA Sequencing Data*. Mississippi State University, May 13, 2025. DOI: 10.54718/PQKJ9866. URL: <https://scholarsjunction.msstate.edu/data-capstone/3/> (visited on 08/10/2026).
- [16] Paolo Tieri et al. “Network Inference and Reconstruction in Bioinformatics”. In: *Encyclopedia of Bioinformatics and Computational Biology*. Elsevier, 2019, pp. 805–813. ISBN: 978-0-12-811432-2. DOI: 10.1016/B978-0-12-809633-8.20290-2. URL: <https://linkinghub.elsevier.com/retrieve/pii/B9780128096338202902> (visited on 08/12/2026).
- [17] Sipko Van Dam et al. “Gene Co-Expression Analysis for Functional Classification and Gene–Disease Predictions”. In: *Briefings in Bioinformatics* (Jan. 10, 2017), bbw139. ISSN: 1467-5463, 1477-4054. DOI: 10.1093/bib/bbw139. URL: <https://academic.oup.com/bib/article-lookup/doi/10.1093/bib/bbw139> (visited on 09/14/2026).
- [18] Petar Veličković et al. *Graph Attention Networks*. Version 3. 2017. DOI: 10.48550/ARXIV.1710.10903. URL: <https://arxiv.org/abs/1710.10903> (visited on 09/11/2026). Pre-published.
- [19] Xiaohan Xing et al. “Multi-Level Attention Graph Neural Network Based on Co-Expression Gene Modules for Disease Diagnosis and Prognosis”. In: *Bioinformatics* 38.8 (Apr. 12, 2022). Ed. by Inanc Birol, pp. 2178–2186. ISSN: 1367-4803, 1367-4811. DOI: 10.1093/bioinformatics/btac088. URL: <https://academic.oup.com/bioinformatics/article/38/8/2178/6528315> (visited on 08/13/2026).